A MODULE FLAT OVER A IS FLAT OVER B WHEN B TENSOR OVER A B MAPS FLATLY ONTO B

THE STACKS PROJECT AUTHORS

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Editorial note. The lead-in sentence at source lines 30198-30199, recording that an etale ring map is weakly etale and announcing that a simple but key property follows, lies before the counted statement and is outside the counted spans.

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Allowed standard prerequisites (editorial).

- A module N is flat over A if and only if tensoring with N over A is an exact functor on the category of A-modules. used at proof lines 30209-30214, where exactness of the functor sending N' to N tensor_A N' is deduced from flatness of N over A. Produces an exact functor out of the category of B-modules as the first half of the composite that has to be shown exact.
- Extension of scalars along a flat ring map is exact, so tensoring with B over B tensor_A B preserves exactness. used at proof lines 30215-30218, where flatness of B tensor_A B to B is applied. Keeps exactness when the functor is composed with the change of scalars back to B.
- A composite of the two functors is naturally isomorphic to tensoring with N over B, so N is flat over B. used at proof lines 30217-30220, where (N tensor_A N') tensor over B tensor_A B B is identified with N tensor_B N'. Identifies the exact functor constructed with the tensor functor of N over B, which is the conclusion.

Lemma 0.2. *Let $A \to B$ be a ring map such that $B \otimes_A B \to B$ is flat. Let N be a B-module. If N is flat as an A-module, then N is flat as a B-module.*

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Proof. Assume N is a flat as an A-module. Then the functor

$$\text{Mod}_B \longrightarrow \text{Mod}_{B \otimes_A B}, \quad N' \mapsto N \otimes_A N'$$

is exact. As $B \otimes_A B \to B$ is flat we conclude that the functor

$$\text{Mod}_B \longrightarrow \text{Mod}_B, \quad N' \mapsto (N \otimes_A N') \otimes_{B \otimes_A B} B = N \otimes_B N'$$

is exact, hence N is flat over B. $\square$